Python Assignment
Topic-3: Core Python Concepts

Q1. Understanding data types: integers, floats, strings, lists, tuples, dictionaries, sets.

Ans.

Data types tell us what kind of value a variable holds in Python. Integer is used for whole numbers like 5 or 100. Float is used for decimal numbers like 10.5. String is used to store text like names. List is used to store multiple values and it can be changed. Tuple is similar to list but it cannot be changed. Dictionary stores data in key and value form. Set stores unique values only. These data types help in storing and managing data properly in programs.

Example:

```
a = 10
b = 10.5
name = "Anshu"
list1 = [1,2,3]
tuple1 = (4,5,6)
dict1 = {"name":"Anshu"}
```

Q2. Python variables and memory allocation.

Ans.

Variables are used to store data in Python. When we assign a value to a variable, Python automatically stores it in memory. We do not need to define the type of variable. Python decides the type based on the value. Variables act like containers that hold data. Python uses dynamic typing, so a variable can change its type later. Memory management is handled by Python itself. This makes programming easier and faster for beginners.

Example:

```
x = 10
x = "Anshu"
print(x)
```

Q3. Python operators: arithmetic, comparison, logical, bitwise.

Ans.

Operators are used to perform operations on values. Arithmetic operators are used for calculations like addition, subtraction, multiplication, and division. Comparison operators are used to compare values and return true or false. Logical operators are used to combine conditions like and, or, and not. Bitwise operators work on binary values. These operators help in building logic in programs. They are very important for decision making and calculations.

Example:

```
a = 15
b = 5
print(a + b)
print(a > b)
print(a > 10 and b < 10)
```